

mathematical formula

mathematical_formula — Formula 1

User Reliability with Confidence Adjustment

Final Formula

$$R_i^*(t) = () (1 - e^{\{-k(_i(t) + _i(t))\}})$$

Interpretation

This formula computes the effective reliability of user i at time t , combining:

- correctness history (Bayesian estimate)
- confidence based on amount of evidence

Components Breakdown

1. Bayesian Reliability Estimate

$$R_i(t) =$$

Represents:

Probability that user i provides correct information.

2. Confidence Term

$$\text{Conf}_i(t) = 1 - e^{\{-k(_i(t) + _i(t))\}}$$

Represents:

How much evidence we have about the user.

3. Final Reliability

$$R_i^*(t) = R_i(t) \text{Conf}_i(t)$$

Ensures:

- new users $\rightarrow$ low influence
- experienced users $\rightarrow$ higher influence

Variable Definitions

User Evidence Variables

- $_i(t)$: time-decayed count of correct actions
- $_i(t)$: time-decayed count of incorrect actions

Decay-Based Computation

$$i(t) = \{k\} e^{-r (t - t_k)} ()$$

$$i(t) = \{k\} e^{-r (t - t_k)} ()$$

Other Variables

- k : confidence growth rate (hyperparameter)
- t_k : timestamp of interaction
- r : time decay constant

Properties

- $R_i^*(t)$
- monotonically increases with correct evidence
- penalizes low sample size
- adapts over time

Why This Formulation Is Correct

This avoids:

- overconfidence from few interactions
- static reputation issues
- dominance of outdated behavior

Usage in System

This value feeds into:

$$w_i(t) = T_i(t) (1 - \text{Anom}_i(t)) \text{Exp}_i(t)$$

Pseudo Insight

Reliability = correctness × confidence

mathematical_formula — Formula 2

User Experience Score

Final Formula

$$\text{Exp}_i(t) = \{(1 + E_{\{\}})\}$$

Interpretation

This formula computes the experience level of user i at time t , representing:

how much meaningful participation the user has accumulated over time.

It ensures:

- diminishing returns (prevents domination by extremely active users)
- bounded output in $[0,1]$
- robustness to spam or burst activity

Core Quantity

$$E_i(t) = \sum_k \{q_k\} e^{-\lambda_E (t - t_k)} q_k$$

Component Breakdown

1. Activity Set

- $\mathcal{A}_i$: set of all actions performed by user i

Each action k could be:

- vote
- post
- interaction

2. Time Decay Term

$$e^{-\lambda_E (t - t_k)}$$

- reduces importance of old actions
- ensures recent activity matters more

3. Quality Weight

- q_k : quality score of action k

Typical choices:

- $q_k = 1$ for valid interaction
- or: $q_k = ()$

4. Raw Experience

$$E_i(t)$$

Represents:

total time-weighted, quality-adjusted activity of user i

5. Log Normalization

$$\text{Exp}_i(t) =$$

Variable Definitions

- $E_i(t)$: raw experience accumulation
- $E_{\max}$: normalization constant (maximum expected experience)
- λ_E : decay rate for experience

- t_k : timestamp of action
- q_k : quality weight

Properties

- $Exp_i(t)$
- monotonic with respect to activity
- diminishing returns due to log scaling
- resistant to spam bursts

Why Log Scaling Is Necessary

Without log:

- highly active users dominate
- system becomes biased

With log:

- early experience matters more
- later growth slows

Behavior Summary

User Type Behavior New user Exp_i Moderate user grows quickly Heavy user saturates near 1

Usage in System

Feeds into:

$$w_i(t) = T_i(t) (1 - Anom_i(t)) Exp_i(t)$$

Pseudo Insight

Experience = time-decayed activity with diminishing returns

mathematical_formula — Formula 3

User Anomaly Score

Final Formula

$$Anom_i(t) = (1 - \alpha) Anom_i^{\text{rule}}(t) + \alpha Anom_i^{\text{ML}}(t)$$

Rule-Based Core

$$Anom_i^{\text{rule}}(t) = 1 - \left(\prod_{m=1}^5 \alpha_m D_m(i, t) \right)$$

ML Component

$$Anom_i^{\text{ML}}(t) = (f(x_i(t)))$$

Interpretation

This formula measures:

how much user i 's behavior deviates from expected human behavior

It combines:

- interpretable rule-based deviations
- learned behavioral patterns (ML)

Component Breakdown

1. Aggregation Mechanism

$$1 - (-x)$$

Ensures:

- output bounded in $[0,1]$
- smooth growth
- no hard thresholds

2. Deviation Components $D_m(i,t)$

1. Temporal Burst Deviation

$$D_1(i,t) =$$

Variables

- $V_i^{\text{window}}(t)$: activity count in recent window
- $\bar{V}_i^{\text{baseline}}(t)$: long-term average activity
- ϵ : small constant

Meaning

Detects:

sudden spikes in activity

2. Behavioral Entropy Deviation

$$H_i(t) = - \sum_a p_i(a,t) \log p_i(a,t)$$

$$D_2(i,t) = 1 -$$

Variables

- $\mathcal{A}$: set of possible actions
- $p_i(a,t)$: probability of action a

Meaning

- low entropy $\rightarrow$ repetitive behavior $\rightarrow$ suspicious

3. Consensus Deviation

$$D_3(i,t) = 1 - \{k\} (s_k = y_{j_k})$$

Variables

- s_k : user vote
- y_{j_k} : final truth of post
- $N_i(t)$: total interactions

Meaning

- persistent disagreement $\rightarrow$ suspicious

4. Coordination Similarity

$$D_4(i,t) = \sum_j (i) \text{Sim}(i,j,t)$$

Similarity Function

$$\text{Sim}(i,j,t) = \{k \{ij\}\} (s_{i,k} = s_{j,k}) (-)$$

Meaning

Detects:

coordinated users acting together

5. Location Inconsistency

$$D_5(i,t) =$$

Meaning

- high value $\rightarrow$ unrealistic movement $\rightarrow$ spoofing

3. ML Feature Vector

$$x_i(t) = [r_{\text{rate}}, r_{\text{timing}}, r_{\text{nav}}, r_{\text{div}}, r_{\text{ip}}, r_{\text{fp}}]$$

ML Output

$$\text{Anom}_i^{\text{ML}}(t) =$$

4. Variable Definitions

- α_m : importance weights for each deviation
- β : blending factor between rule-based and ML
- γ : time alignment parameter
- (i) : neighboring users
- $\{ij\}$: shared posts
- y_j : ground truth label

5. Properties

- $Anom_i(t)$
- smooth and differentiable
- robust to noise
- captures multiple attack patterns

6. Behavior Interpretation

Value Meaning ~0 normal behavior ~0.5 suspicious ~1 highly anomalous

7. Why This Formulation Is Strong

- no single signal dominates
- resistant to evasion
- interpretable + extensible
- compatible with real-time updates

8. Usage in System

$$w_i(t) = T_i(t) (1 - Anom_i(t)) Exp_i(t)$$

Pseudo Insight

Anomaly = aggregated deviation from human behavior

Proceeding with the next dependent component.

mathematical_formula — Formula 4

Graph-Based Trust Propagation

Final Formula

$$T^{\{(k+1)\}}(t) = _g , (t), T^{\{(k)\}}(t) + (1 - _g), R^*(t)$$

Converged Trust

$$T(t) = _k \} T^{\{(k)\}}(t)$$

Interpretation

This formula computes the network-adjusted trust score of users, where:

trust is influenced both by individual reliability and by neighboring users in the interaction graph.

Component Breakdown

1. Trust Vector

$T(t) =$
Meaning

- $T_i(t)$: final trust score of user i

2. Base Reliability Vector

$$R^*(t) =$$

Meaning

- reliability computed from Formula 1

3. Normalized Adjacency Matrix

$$_{\{ij\}}(t) =$$

Meaning

- row-normalized matrix
- each row sums to 1

Interpretation

$$_{\{ij\}}(t)$$

represents:

fraction of influence user i assigns to user j

4. Iterative Propagation Term

$$(t), T^{\{(k)\}}(t)$$

Meaning

- aggregates trust from neighbors
- weighted by similarity

5. Damping / Anchor Term

$$(1 - \alpha_g), R^*(t)$$

Meaning

- prevents drift
- ensures trust remains grounded in observed behavior

6. Mixing Parameter

- $\alpha_g (0,1)$

Interpretation

Value Behavior small trust mostly individual large trust heavily network-influenced

Initialization

$$T^{\{0\}}(t) = R^*(t)$$

Convergence Condition

$$|T^{\{k+1\}}(t) - T^{\{k\}}(t)| <$$

Properties

- $T_i(t)$
- converges under damping
- stable against noise
- incorporates global structure

Why This Formulation Works

This formulation ensures:

- trusted users reinforce each other
- isolated malicious clusters cannot dominate
- reliability is corrected using relational context

Behavior Interpretation

Scenario Outcome user connected to trusted users trust increases user in suspicious cluster trust suppressed isolated user trust $\approx$ reliability

Usage in System

$$w_i(t) = T_i(t) (1 - \text{Anom}_i(t)) \text{Exp}_i(t)$$

Pseudo Insight

Trust = reliability refined by network structure

mathematical_formula — Formula 5

Final User Voting Weight

Final Formula

$$w_i(t) = T_i(t);;(1 - \text{Anom}_i(t));;\text{Exp}_i(t)$$

Interpretation

This formula defines the influence weight of user i at time t , which determines:

how strongly the user's actions affect post credibility, propagation, and system decisions.

It integrates three independent dimensions:

- trustworthiness (graph-adjusted reliability)
- behavioral integrity (absence of anomaly)
- evidence strength (experience level)

Component Breakdown

1. Trust Component

$T_i(t)$

Meaning

- derived from Formula 4 (graph propagation)
- captures:
 - individual correctness
 - network influence

Role

- amplifies users embedded in trustworthy regions
- suppresses users connected to suspicious clusters

2. Anomaly Suppression

$1 - \text{Anom}_i(t)$

Meaning

- derived from Formula 3 (anomaly score)
- converts anomaly into a penalty factor

Behavior

$\text{Anom}_i(t)$ Effect 0 no penalty 0.5 weight halved 1 weight nullified

Role

- prevents adversarial users from influencing system
- ensures abnormal behavior reduces impact

3. Experience Component

$\text{Exp}_i(t)$

Meaning

- derived from Formula 2 (experience score)
- reflects accumulated activity

Role

- prevents new users from having large influence
- rewards consistent participation

Combined Effect

The product form ensures:

$$w_i(t) = 0$$

Interpretation

A user contributes meaningfully only if:

- they are trusted
- they behave normally
- they have sufficient experience

Variable Definitions

- $T_i(t)$: graph-based trust score
- $Anom_i(t)$: anomaly score
- $Exp_i(t)$: experience score

All values lie in:

$[0,1]$

Properties

- $w_i(t)$
- multiplicative gating mechanism
- robust to manipulation
- dynamically updates over time

Behavior Summary

User Type Outcome
trusted + normal + experienced high weight anomalous user strongly suppressed
new user low weight isolated unreliable user low weight

Why Multiplicative Form Is Correct

This avoids:

- additive leakage (where one strong signal dominates)
- partial trust exploitation

Instead enforces:

all conditions must be satisfied simultaneously

Usage in System

This is used in:

Interaction aggregation

$N_j(t), S_j(t)$

Credibility computation

$C_{\{Bayes\}}(j,t)$

Variance

$Var_j(t)$

Pseudo Insight

Influence = trust × integrity × experience

Proceeding to the next post-level foundation.

mathematical_formula — Formula 6

Effective Interaction Mass

Final Formula

$N_j(t) = \sum_k w_{i_k}(t_k); e^{-(t - t_k)}$

Interpretation

This formula computes the effective amount of evidence supporting post j at time t .

It represents:

the total trust-weighted, time-decayed interaction mass accumulated for the post.

Component Breakdown

1. Interaction Set

$\mathcal{I}_j$

Meaning

- set of all interactions associated with post j
- each interaction indexed by k

2. User Weight

$w_{i_k}(t_k)$

Meaning

- influence of the user who performed interaction k
- computed from Formula 5

Role

- trusted users contribute more
- anomalous or inexperienced users contribute less

3. Time Decay Factor

$$e^{-(t - t_k)}$$

Variables

- t_k : time of interaction k
- t : current time
- $\lambda > 0$: decay rate

Meaning

- reduces influence of older interactions
- ensures system reacts to recent evidence

4. Summation

$$\sum_{j \in \mathcal{I}_k} w_{i_k}(t_k) e^{-(t - t_k)}$$

Meaning

- aggregates contributions from all interactions

Variable Definitions

- $N_j(t)$: effective interaction mass
- $w_{i_k}(t_k)$: user weight
- λ : temporal decay parameter
- t_k : timestamp of interaction
- $\mathcal{I}_k$: interaction set

Properties

- $N_j(t)$
- increases with more interactions
- decreases over time if no new activity
- sensitive to both quantity and quality

Behavior Interpretation

Scenario Effect on $N_j(t)$ many trusted users high value many low-trust users limited growth old interactions decay over time recent burst sharp increase

Why This Formulation Is Necessary

Without weighting:

- all users equal $\rightarrow$ vulnerable to spam

Without decay:

- old data dominates $\rightarrow$ stale system

With both:

system remains adaptive and robust

Usage in System

Used in:

- credibility estimation
- propagation threshold: $N_j(t) \geq N_{\min}$
- urgency computation (via rate)

Pseudo Insight

Evidence = sum of trusted, recent interactions

Proceeding to the next core post-level quantity.

mathematical_formula — Formula 7

Weighted Signal Aggregation

Final Formula

$$S_j(t) = \sum_k w_{i_k}(t_k) s_k e^{-(t - t_k)}$$

Positive and Negative Decomposition

$$S_j^+(t) = \sum_{k: s_k = +1} w_{i_k}(t_k) e^{-(t - t_k)}$$

$$S_j^-(t) = \sum_{k: s_k = -1} w_{i_k}(t_k) e^{-(t - t_k)}$$

$$S_j(t) = S_j^+(t) - S_j^-(t)$$

Interpretation

This formula computes the net directional evidence for post j at time t .

It represents:

the balance between trusted positive support and trusted negative opposition, weighted by recency.

Component Breakdown

1. Interaction Set

$\mathcal{I}_j$

- all interactions associated with post j

2. User Weight

$w_{i_k}(t_k)$

- from Formula 5
- encodes:

- trust
- anomaly penalty
- experience

3. Vote Signal

$s_k \{-1, +1\}$

Meaning

- +1: supports the claim
- -1: opposes the claim

4. Time Decay

$e^{-(t - t_k)}$

- reduces influence of older interactions
- aligns with Formula 6

Variable Definitions

- $S_j(t)$: net signal
- $S_j^{+}(t)$: positive support mass
- $S_j^{-}(t)$: negative support mass
- s_k : vote value
- $w_{\{i_k\}}(t_k)$: user weight
- λ : decay rate

Properties

- $S_j(t)$
- sign indicates direction of consensus
- magnitude indicates strength of agreement

Behavior Interpretation

Scenario Outcome strong agreement (+) S_j strong disagreement (-) S_j mixed opinions S_j

Why Decomposition Matters

Separating:

$S_j^{+}(t), ; S_j^{-}(t)$

is essential for:

- Bayesian credibility computation
- variance calculation
- understanding polarization

Relation to Interaction Mass

From Formula 6:

$$N_j(t) = S_j^{+}(t) + S_j^{-}(t)$$

Usage in System

Feeds into:

$$C_{\{Bayes\}}(j,t)$$

and

$$Var_j(t)$$

Pseudo Insight

Signal = trusted, time-weighted direction of opinion

mathematical_formula — Formula 8

Bayesian Credibility Estimation

Final Formula

$$C_{\{Bayes\}}(j,t) =$$

Interpretation

This formula computes the probabilistic credibility of post j at time t , based on:

trusted positive and negative evidence, adjusted with prior belief.

It represents:

$$P()$$

Component Breakdown

1. Positive Evidence

$$S_j^{+}(t)$$

- from Formula 7
- represents trusted support for the post

2. Negative Evidence

$$S_j^{-}(t)$$

- from Formula 7
- represents trusted opposition

3. Prior Parameters

$$\alpha_0, \beta_0 > 0$$

Meaning

- α_0 : prior positive belief
- β_0 : prior negative belief

Role

- stabilizes early-stage estimates
- prevents extreme values with low data

4. Denominator

$$S_j^+(t) + S_j^-(t) + \alpha_0 + \beta_0$$

Meaning

- total effective evidence
- ensures normalization

Variable Definitions

- $C_{\text{Bayes}}(j,t)$: credibility estimate
- $S_j^+(t)$: weighted positive signal
- $S_j^-(t)$: weighted negative signal
- α_0, β_0 : prior constants

Properties

- bounded in $[0,1]$
- smooth update as evidence accumulates
- robust in low-data regimes
- interpretable as probability

Behavior Interpretation

Scenario Outcome strong positive evidence C_{Bayes} strong negative evidence C_{Bayes} balanced evidence C_{Bayes} no evidence $C_{\text{Bayes}} =$

Why Priors Are Necessary

Without priors:

- early votes dominate
- instability in low interaction phase

With priors:

system starts with controlled uncertainty

Relation to Interaction Mass

$$S_{j^+}(t) + S_{j^-}(t) = N_j(t)$$

Usage in System

Feeds into final credibility:

$$C_j(t) = (1-\alpha)C_{\text{Bayes}} + C_{\text{ML}} + C_{\text{memory}}$$

Pseudo Insight

Credibility = ratio of trusted positive evidence to total evidence

mathematical_formula — Formula 9

Credibility Variance (Stability Measure)

Final Formula

$$\text{Var}_j(t) =$$

Interpretation

This formula measures the degree of disagreement among users regarding post j at time t .

It represents:

how much individual opinions deviate from the current credibility estimate.

Component Breakdown

1. Interaction Set

I_j

- all interactions associated with post j

2. User Weight

$w_{i_k}(t_k)$

- from Formula 5
- ensures trusted users influence variance more

3. Vote Signal

$s_k \in \{-1, +1\}$

4. Credibility Mean

$C_j(t)$

- final credibility of post

- (can be C_{Bayes} or full combined C_j)

5. Squared Deviation

$$(s_k - C_j(t))^2$$

Meaning

- measures how far a vote deviates from consensus

6. Time Decay

$$e^{-(t - t_k)}$$

- reduces influence of older interactions

7. Normalization

$$w_{\{i_k\}}(t_k); e^{-(t - t_k)} = N_j(t)$$

Variable Definitions

- $\text{Var}_j(t)$: variance of credibility
- $C_j(t)$: current credibility estimate
- s_k : vote value
- $w_{\{i_k\}}(t_k)$: user weight
- λ : decay parameter

Properties

- $\text{Var}_j(t)$
- bounded (typically within a finite range due to $s_k \in \{-1, 1\}$)
- decreases as consensus increases

Behavior Interpretation

Scenario Outcome strong agreement low variance mixed opinions high variance polarized votes high variance

Why This Formulation Is Necessary

Credibility alone is insufficient.

Two cases:

- $C_j = 0.8$ with strong agreement $\rightarrow$ reliable
- $C_j = 0.8$ with disagreement $\rightarrow$ unstable

Variance distinguishes these.

System Role

Used in:

Propagation condition

$\text{Var}_j(t)^2$

Alert condition

$\text{Var}_j(t)^2$

Key Insight

Variance captures confidence in credibility.

Pseudo Insight

Variance = disagreement weighted by trust and recency

mathematical_formula — Formula 10

Final Post Credibility (Multi-Signal Integration)

Final Formula

$C_j(t) = (1 - \alpha) \cdot C_{\text{Bayes}}(j,t) + \alpha \cdot C_{\text{ML}}(j,t) + \beta \cdot C_{\text{memory}}(j,t)$

Interpretation

This formula computes the final credibility of post j at time t , by combining:

- community-based evidence
- machine learning prediction
- historical knowledge (memory)

It represents:

the system's best estimate of whether the post is credible.

Component Breakdown

1. Bayesian Credibility

$C_{\text{Bayes}}(j,t)$

Source

- from Formula 8

Role

- primary signal
- derived from trusted user interactions

2. ML-Based Credibility

$C_{\text{ML}}(j,t)$

Meaning

- probability from content model
- based on text semantics

Role

- early-stage signal (before many votes)
- captures linguistic cues

3. Memory-Based Credibility

$C_{\{\text{memory}\}}(j,t)$

Definition

$C_{\{\text{memory}\}}(j,t) =$

Variables

- (j) : set of similar past posts
- $\text{Sim}(j,k)$: similarity between posts

Role

- leverages historical patterns
- reinforces known truths or misinformation

4. Weight Parameters

- α : ML contribution weight
- β : memory contribution weight

Constraint

$0 \leq \alpha + \beta \leq 1$

Meaning

- remaining weight: $1 - \alpha - \beta$ - assigned to Bayesian signal

Variable Definitions

- $C_j(t)$: final credibility
- $C_{\{\text{Bayes}\}}$: interaction-based credibility
- $C_{\{\text{ML}\}}$: content-based probability
- $C_{\{\text{memory}\}}$: similarity-based estimate

Properties

- bounded in $[0,1]$
- interpretable linear combination

- stable under noisy signals
- extensible (new terms can be added)

Behavior Interpretation

Scenario Dominant Signal early stage ML many interactions Bayesian repeated claim memory

Why This Formulation Is Correct

Avoids:

- ML-only decisions
- crowd-only bias
- ignoring historical knowledge

Instead ensures:

balanced multi-source credibility estimation

Usage in System

Used in:

- propagation condition
- alert decision
- variance computation (as mean)

Pseudo Insight

Credibility = evidence (users) + semantics (ML) + history (memory)

mathematical_formula — Formula 11

User–Post Proximity Function

Final Formula

$$\text{Prox}(u,j,t) = L_u(t); ; (-)$$

Interpretation

This formula computes the relevance of post j to user u at time t , based on:

- geographic distance
- confidence in user's location

It represents:

how appropriate it is to show or alert this post to a specific user.

Component Breakdown

1. Location Confidence

$L_u(t)$

Meaning

- trust in the user's reported location
- derived from spatial consistency signals

Role

- reduces influence of unreliable or spoofed locations

2. Distance Function

$d(u,j) = d(l_u(t), l_j)$

Variables

- $l_u(t)$: location of user u
- l_j : location of post j

Meaning

- geographic distance (e.g., haversine distance)

3. Gaussian Distance Decay

(-)

Variables

- $\alpha_p > 0$: spatial decay parameter

Meaning

- nearby posts $\rightarrow$ high relevance
- distant posts $\rightarrow$ rapidly decreasing relevance

Variable Definitions

- $\text{Prox}(u,j,t)$: proximity score
- $L_u(t)$: location confidence
- $d(u,j)$: geographic distance
- α_p : spatial scale

Properties

- bounded in $[0,1]$
- smooth decay with distance
- penalizes unreliable locations
- symmetric with respect to distance

Behavior Interpretation

Distance Prox very close near 1 moderate decreases smoothly far away near 0

Effect of Location Confidence

$L_u(t)$ Effect 1 full proximity 0.5 reduced relevance 0 no relevance

Why Gaussian Decay Is Used

Compared to linear decay:

- smoother transition
- avoids sharp cutoffs
- better reflects real-world spatial relevance

System Role

Used in:

Alert decision

$\text{Prox}(u,j,t) _p$

Feed ranking

- higher proximity $\rightarrow$ higher priority

Key Insight

=

Pseudo Insight

Proximity = trusted closeness in space

mathematical_formula — Formula 12

Post Urgency Score

Final Formula

$$U_j(t) = 1 - (- (_1 K_j + _2 \text{Cat_}j + _3 V_j(t)))$$

Interpretation

This formula computes the urgency level of post j at time t , representing:

how time-sensitive and action-relevant the information is.

It combines:

- textual urgency signals
- semantic category importance
- real-time activity velocity

Component Breakdown

1. Keyword-Based Signal

$$K_j = \sum \{w \mid p_j\} (w)$$

Variables

- p_j : set of words in post j
- (w) : importance score of word w

Meaning

- captures urgency-related terms (e.g., emergency indicators)

2. Category-Based Signal

$$Cat_j$$

Meaning

- output of content classification
- represents importance level of category

Example

Category Value emergency high general low

3. Velocity-Based Signal

Step 1: Interaction Rate

$$rate_j(t) =$$

Step 2: Normalization

$$V_j(t) = 1 - (-)$$

Meaning

- detects rapidly evolving events
- captures bursts of activity

4. Weighted Aggregation

$$_1 K_j + _2 Cat_j + _3 V_j(t)$$

Constraints

$$_1 + _2 + _3 = 1$$

Meaning

- balances different urgency signals

5. Final Bounding Function

1 - (-x)

Purpose

- ensures output in [0,1]
- smooth response to input

Variable Definitions

- $U_j(t)$: urgency score
- K_j : keyword urgency
- Cat_j : category importance
- $V_j(t)$: activity velocity
- $_1, _2, _3$: weights
- t : time window
- $rate_{\{baseline\}}$: normalization constant

Properties

- bounded in [0,1]
- increases with urgency signals
- robust to single-signal manipulation
- responsive to real-time changes

Behavior Interpretation

Scenario $U_j(t)$ urgent keywords + burst high normal content low high activity spike increases

Why Multi-Signal Is Necessary

Single signals can be manipulated:

- keywords $\rightarrow$ spammed
- velocity $\rightarrow$ bot attack

Combined:

system becomes robust

System Role

Used in:

Alert condition

$C_j(t) U_j(t) _{\{alert\}}$

Key Insight

= +

Pseudo Insight

Urgency = how important and how fast the situation is evolving

mathematical_formula — Formula 13

Propagation Condition (Controlled Expansion)

Final Formula

$$\text{Expand}(j,t) = (C_j(t) \;; N_j(t) \geq N_{\min} \;; \text{Var_}j(t) \leq \text{Var_}j^2 \;; (t - t_{\text{create},j}) \geq T_{\min} \;; \{L\}_j(t) \geq L_{\min})$$

Interpretation

This formula determines whether post j is allowed to expand its visibility at time t .

It represents:

a strict multi-condition gate ensuring that content spreads only when sufficiently credible, supported, stable, and trustworthy.

Component Breakdown

1. Indicator Function

$()$

Meaning

- returns:
 - 1 → expand
 - 0 → do not expand

2. Credibility Constraint

$C_j(t)$

Meaning

- ensures post has sufficient credibility

Variable

- : credibility threshold

3. Evidence Constraint

$N_j(t) \geq N_{\min}$

Meaning

- ensures enough interaction evidence

Variable

- $N_{\min}$: minimum required interaction mass

4. Stability Constraint

$\text{Var}_j(t)^2$

Meaning

- ensures consensus among users

Variable

- 2 : variance threshold

5. Temporal Stability Constraint

$(t - t_{\text{create},j}) T_{\min}$

Meaning

- ensures post has existed long enough for evaluation

Variable

- $T_{\min}$: minimum stabilization time

6. Spatial Trust Constraint

$\{L\}_j(t) L_{\min}$

Definition

$\{L\}_j(t) =$

Meaning

- average location trust of contributing users

Variable

- $L_{\min}$: minimum spatial trust

Variable Definitions

- $C_j(t)$: final credibility
- $N_j(t)$: interaction mass
- $\text{Var}_j(t)$: credibility variance
- $t_{\text{create},j}$: post creation time
- $\{L\}_j(t)$: aggregated location confidence

Properties

- binary decision
- multi-dimensional constraint

- prevents premature propagation
- robust to manipulation

Behavior Interpretation

Condition Fails Result low credibility blocked insufficient data blocked disagreement blocked too early
delayed unreliable location blocked

Why Multi-Constraint Is Necessary

Single-condition systems fail because:

- high votes $\neq$ truth
- high activity $\neq$ reliability

This formulation ensures:

propagation requires consistent agreement across all dimensions

System Role

Used to:

- control propagation radius
- determine expansion step

Key Insight

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Pseudo Insight

Content spreads only when it earns the right to spread

mathematical_formula — Formula 14

Alert Decision Function

Final Formula

$$\text{Alert}(u, j, t) = (\text{Prox}(u, j, t) \cdot p \;; \; C_j(t) U_j(t) \{ \text{alert} \} \;; \; \text{Var}_j(t) ^2 \;; \; \text{RateLimited}(u, t))$$

Interpretation

This formula determines whether user u should receive an alert for post j at time t .

It represents:

a strict, user-specific decision that ensures alerts are relevant, credible, urgent, stable, and not excessive.

Component Breakdown

1. Indicator Function

()

Meaning

- returns:
 - 1 → send alert
 - 0 → do not send

2. Proximity Constraint

$\text{Prox}(u,j,t) \text{ } _p$

Meaning

- ensures the post is geographically relevant

Variables

- $\text{Prox}(u,j,t)$: from Formula 11
- $_p$: proximity threshold

3. Credibility–Urgency Constraint

$C_j(t)U_j(t) \text{ } _{\text{alert}}$

Meaning

- ensures the post is both:
 - credible
 - urgent

Variables

- $C_j(t)$: from Formula 10
- $U_j(t)$: from Formula 12
- $_{\text{alert}}$: alert threshold

4. Stability Constraint

$\text{Var_j}(t) \text{ } ^2$

Meaning

- ensures the information is not controversial or unstable

5. Rate Limiting Constraint

$\text{RateLimited}(u,t)$

Definition

$\text{RateLimited}(u,t) = (N_ {\text{alerts}}(u, t-t, t) R_ {\text{max}})$

Variables

- $N_{\text{alerts}}(u, t-t, t)$: alerts received in window
- R_{max} : maximum allowed alerts

Meaning

- prevents alert fatigue

Variable Definitions

- $\text{Prox}(u, j, t)$: spatial relevance
- $C_j(t)$: credibility
- $U_j(t)$: urgency
- $\text{Var}_j(t)$: variance
- $_p$: proximity threshold
- $_{\text{alert}}$: alert threshold
- R_{max} : rate limit

Properties

- binary decision
- user-specific
- multi-dimensional gating
- robust against spam and noise

Behavior Interpretation

Condition Fails Result far away no alert low credibility no alert low urgency no alert high disagreement no alert too many alerts suppressed

Why Multiplicative Credibility–Urgency

$C_j(t)U_j(t)$

ensures:

- urgent but false $\rightarrow$ blocked
- true but non-urgent $\rightarrow$ no alert
- only credible AND urgent $\rightarrow$ allowed

System Role

- triggers notifications
- defines real-time system output
- directly affects user experience

Key Insight

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Pseudo Insight

Alert only when it matters, is true, and affects the user

Status

At this point:

- all major formulas in your system are defined
- dependencies are complete
- no variable remains undefined